

COURSE DESCRIPTION			
Program	Printing Technology		
Course Title	Fundamentals of Printing Science		
Course Code	2101	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	4:0:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

This course is designed to give a clear and theoretical foundation in printing technology. It helps students understand how color is seen and reproduced, how digital files and images are prepared for printing, and how tones are controlled to achieve acceptable print quality. The course also introduces the basic characteristics of printing materials and their proper handling, enabling students to work efficiently and confidently in print production environments.

Course Objectives

1	Articulate the fundamental knowledge of color, human visual perception, color theories, and illumination concepts for accurate color understanding and reproduction in printing.
2	To enable the learners with the knowledge and skills required to handle digital content for print production
3	To provide basic knowledge of how images are formed, measured, and reproduced in printing, so they can control tone and quality effectively during print production.
4	To provide basic knowledge of material characteristics used in printing so they can select and handle them properly to achieve smooth operation and acceptable print quality.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Electro magnetic spectrum, Newton's color wheel, Light, Viscosity,	2002A	Engineering Physics for Mechanical, Structural and Industrial Applications	I/II

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Explain the principles of color and human visual perception and apply color harmony, color mixing, and illumination concepts for accurate and consistent color reproduction in printing.	Understand
CO2	Articulate digital materials for printing by applying basic knowledge of image quality, color reproduction, and image input processes used in print production.	Understand
CO3	Explain how images are reproduced with tonal variation and how print quality is evaluated and controlled using basic measurement concepts.	Understand
CO4	Explain how the characteristics of printing substrates and inks influence printability, machine performance, and final print quality	Understand

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	1	1	1	-	-	-	-	1
CO2	3	2	1	1	2	1	-	-	-	-	1
CO3	3	3	1	2	1	1	-	-	-	-	1
CO4	3	2	1	1	1	2	1	1	-	-	1
Total	12	9	3	5	5	5	1	1	-	-	4
Correlation Level	3	2.25	1	1.25	1.25	1.25	1	1	-	-	1

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
4	0	0	0	6	60	4	40	60	100	0	0	0	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Color Science	Color: Definition. Attributes of color - Hue, Saturation, Lightness, Alliance of EM Spectrum, Wavelength of Light and Color, Human Color perception - Elements of eye, Rod and LMS Cone cell vision. Standard illuminant - Definition, Types of illuminants, Color temperature. Spectral power distribution curve Color Schemes: Warm color, Cool color, Monochromatic, Analog and Complementary color. Color wheel. Memory color, Metamerism. Color Theories: Additive and Subtractive color theories. Color theories Vs Color perception and Color reproduction. Maxwell Color Triangle.	15	0
Module II	Digital Imaging	Conventional color separation & Electronic color separation Scanners - Working Principle & Types. Digital image - Pixel and bit depth. Vector file formats and raster file formats. Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF, BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD, PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML.	15	0
Module III	Halftone reproduction	Type of Originals - Line original, Continuous tone originals, Halftone original, Areas of continuous tone image. Halftone screening - AM, FM & Hybrid Screening process. Halftone dots - features and types. Optical Density - Definition, plays a role in printing. Tonal Gradation - Definition. Halftone image - Significance and basics of halftone imaging. Light sensitive material - Film (positive and negative), basics of image formation on film	15	0
Module IV	Paper and Ink Science in Printing Technology	Paper and Board Properties: Define - Paper and Board- Sizes, Classifications and their uses Printability & Runnability Properties. Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain Direction, Porosity, Smoothness. Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding Endurance, Stiffness, Abrasion Resistance. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity.	15	0

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Attributes of color	Distinguish between hue, saturation, and lightness in visual examples.	L	CO1	Understand	1
Electromagnetic Spectrum	Explain the significance of the electromagnetic spectrum in color perception and printing	L	CO1	Understand	1
Visible spectrum	Explain the visible spectrum and relate different wave length to corresponding color sensations relevant to visual perception and printing.	L	CO1	Understand	1
Human Color perception	Describe the structure and function of the human eye and explain how different parts contribute to the perception of color.	L	CO1	Understand	1

Human Color perception	Differentiate between rod and cone cells (LMS) and explain their specific roles in detecting light intensity and color.	L	CO1	Understand	1
Standard illuminant	Importance of standardized lighting condition for consistent color evaluation & purpose of standard illuminants in color measurement	L	CO1	Understand	1
Standard illuminant	Classification & characteristics of commonly used illuminants.	L	CO1	Understand	1
Standard illuminant	Definition of color temperature and its relation to the appearance of light sources.	L	CO1	Understand	1
Spectrum distribution curve	Definition and significance of SPD in describing the energy output of light sources across different wavelengths	L	CO1	Understand	1
Spectrum distribution curve	Comparison of SPD curves for daylight, fluorescent, tungsten, LED, and other common illuminants.	L	CO1	Understand	1
Color Schemes	Definition, characteristics, and applications of warm and cool color groups. Monochromatic Color Scheme, Analogous Color Scheme, Complementary Color Scheme, Memory color and Metamerism. Construction, interpretation, and use of the color wheel & Maxwell's color triangle	L	CO1	Understand	2
Color Theories	Principle of RGB color mixing, characteristics, and applications in displays and imaging. Principle of CMY/CMYK color mixing, characteristics, and applications in printing. Comparison of Additive and Subtractive Color Theories	L	CO1	Understand	3
Module II					
Different File formats	Identify different digital file formats, explain their features and uses - JPEG/JPG, TIFF, PNG, GIF, BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD, PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML	L	CO2	Understand	3
Vector file formats & raster file formats.	Explain the Characteristics & applications of vector and raster file formats	L	CO2	Understand	1
Digital image	Define pixel and bit depth in digital images. Explain the effect of pixel resolution and bit depth on image quality and clarity. Relate pixel and bit depth settings to file size and print quality. Select suitable pixel resolution and bit depth for basic print and digital applications.	L	CO2	Understand	4
Color separation	Describe the basic principles of conventional color separation used in traditional printing workflows. Explain the process of electronic color separation in digital prepress systems. Differentiate between conventional and electronic color separation methods.	L	CO2	Understand	4
Scanners	Working principle, Advantages & Limitations of a) Drum scanner b) Flat bed scanner	L	CO2	Understand	3
Module III					
Type of Originals	Definition of originals in printing, Classification of originals- Line original, Continuous tone originals, Halftone originals- Characteristics of each type Areas of continuous tone image -Identify the different areas of a continuous-tone image such as highlights, mid-tones, and shadows. Explain the tonal characteristics of each area in continuous tone image	L	CO3	Understand	4
Halftone screening	Concept of halftone screening - AM screening, FM screening, Hybrid screening process.	L	CO3	Understand	3
Halftone dots	Definition, Need, features and types.	L	CO3	Understand	1
Halftone imaging	Definition & Need, Conversion of continuous-tone images into halftone images. Explain Role of screening in halftone image formation. Describe Conversion of continuous-tone images into halftone images	L	CO3	Understand	4

Light sensitive material	Explain Role of photographic film in conventional prepress and image reproduction. Describe the Structure and types of Film and basics of Image Formation on Film	L	CO3	Understand	3
Module IV					
Paper and Board Properties	Define and differentiate paper and board based on size, classification, and end-use applications in printing and packaging	L	CO4	Understand	1
Printability and runnability properties	Explain the concepts of printability and runnability properties and their influence on machine performance and print quality	L	CO4	Understand	2
optical properties	Identify and describe the optical properties of paper such as color, gloss, opacity, brightness, whiteness, and metamerism, and their role in visual appearance of printed products.	L	CO4	Understand	2
Structural properties	Analyze the structural properties of paper and board, including density, basis weight, grammage, caliper, bulk, wire and felt sides, moisture content, relative humidity, dimensional stability, grain direction, porosity, and smoothness.	L	CO4	Understand	2
Optical properties	Describe the optical properties of printing inks including color, gloss, opacity, and transparency and their impact on color reproduction	L	CO4	Understand	2
Physical properties	Explain the physical properties of inks such as viscosity, tack, flow, adhesion, length, and scuff resistance in relation to ink transfer and print quality.	L	CO4	Understand	3
Rheological properties	Interpret the rheological behavior of printing inks by distinguishing plastic, pseudo plastic, dilatant, and thixotropic characteristics and explaining viscoelasticity.	L	CO4	Understand	3

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Create small design samples demonstrating: • Warm color scheme • Cool color scheme • Monochromatic scheme • Analogous scheme • Complementary scheme	6
2	Prepare a brief note explaining color and its attributes (hue, saturation, lightness) with real-life examples.	6
3	Compare color appearance under daylight, tungsten, and LED light sources and prepare a report.	6
Module II		
4-5	Prepare a table listing different file formats used in printing and group formats as image, document, layout, color management, and workflow files. Mention their features, advantages, limitations, and typical printing applications.	12
6-7	Explain the role of pixel count and bit depth in print quality.	12
8-9	Draw a simple block diagram showing the working of a scanner.	12
Module III		
10-11	Take a photograph or printed picture and mark highlight, midtone, and shadow areas using a pencil or digital tool. Write a short note explaining how tonal variation appears in different areas.	12
12-13	Identify and explain which screening method is most suitable for different print applications such as newspapers, packaging, and high-quality art prints.	12
Module IV		
14	Prepare a one-page report on "Role of Paper and Ink Properties in Print Quality" with examples.	6
15	Match paper types with suitable printing applications (newspaper, books, packaging, labels).	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK

Sl. No.	Title	Author	Publication
1	Chemistry in Printing, 2nd Edition, 2011	Tulika Das	Barnana Prakashani
2	Understanding Digital Colour, 1995	P. Green	GATF

REFERENCE			
Sl. No.	Title	Author	Publication
1	The Printing Ink Manual, 5th Edition,	R.H. Leach and R.J. Pierce Ed,	Kluwer Academic publisher.
2	A Hand Book for Printing and Packaging, 1997.	Bishwanath Chakravarty	Galgotia Publications Pvt. Ltd
3	Understanding Colour Management	A. Sharma	Thompson Dalmar Learning
4	Heidelberg Expert guide PDF workflow..	Heidelberg	Heidelberg
5	Handbook of printing and production,	Michael Bernard, John Peacock.	Springer Science & Business.
6	Hand book of print Media	Helmut Kipphan	Springer Science & Business.

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://www.colormatters.com	I
2	https://www.britannica.com	II
3	https://www.prepressure.com	III
4	https://www.paperonweb.com	IV

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	

2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60

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